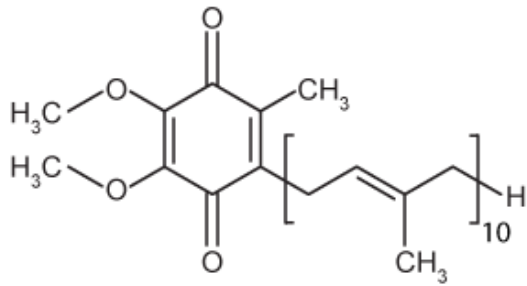
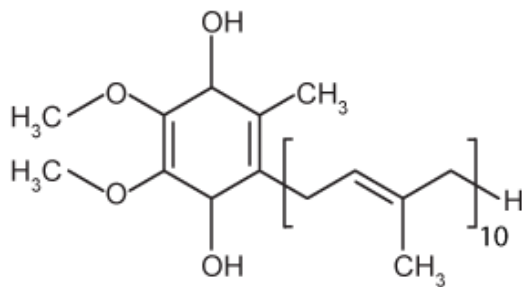


Ubiquinone (shown below) is an important molecule in the electron transport chain, where it receives electrons in complex I originating from a hydride ion in NADH.



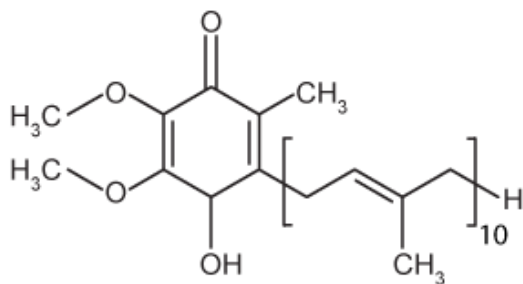
What is the structure of the product formed when two electrons are added to ubiquinone?

☐ A.



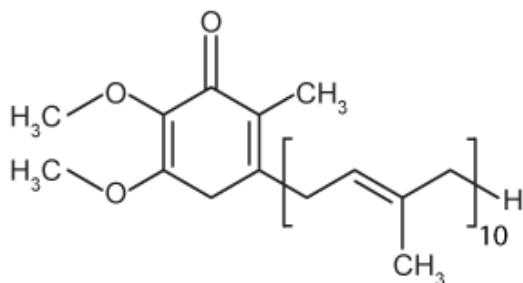
[37%]

☐ B.



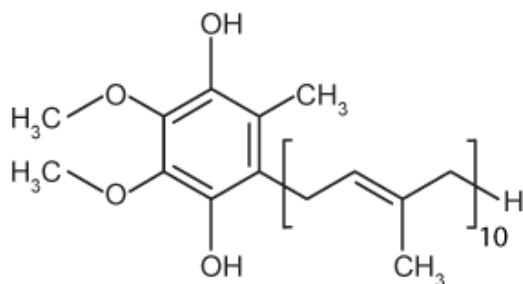
[13%]

☐ C.



[2%]

✓ ☒ D.

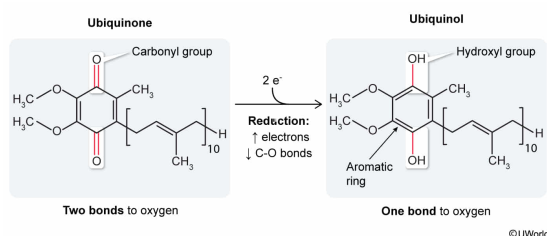


[46%]

Correct

46% answered correctly

Explanation:



Oxidation-reduction (redox) reactions occur when **electrons** are **transferred** from one atom to another. A molecule is oxidized when it loses electrons and the number of **carbon-heteroatom bonds** increases (eg, in a polar covalent bond, the carbon *loses electrons* to an electronegative heteroatom). A molecule is reduced when it gains electrons and the number of carbon-heteroatom bonds decreases. Reduction can also be viewed as an increase in the number of bonds to hydrogen.

Ubiquinone is the fully oxidized form of coenzyme Q, which participates in the electron transport chain in the mitochondrial membrane. The **two electrons** that ubiquinone receives from NADH **reduce** the two carbonyls (C=O) in ubiquinone to hydroxyl (–OH) groups. When ubiquinone gains electrons, the number of **carbon-heteroatom bonds** on each carbonyl carbon **decreases from two to one**. Electrons are also distributed to the six-membered ring, making it **aromatic**. Therefore, the structure of ubiquinol contains two hydroxyl groups on an aromatic ring.

(Choice A) The ring should be an aromatic ring rather than the diene present in the six-membered ring of ubiquinone. The structure shown here would only arise if *four* electrons (and four protons) were added to ubiquinone; however, the addition of the two extra electrons would break the stable **aromatic ring** and break the conjugation of the system by converting the two hydroxyl carbons to sp^3 hybridization.

(Choice B) Both carbonyl carbons should be reduced to hydroxyls (–OH), and the ring should be an aromatic ring rather than the diene present in ubiquinone.

(Choice C) This structure shows the over-reduction of one carbonyl carbon with complete removal of the oxygen atom. Both carbonyl carbons should be reduced to hydroxyls (–OH), and the ring should be an aromatic ring.

Educational objective:

Oxidation-reduction reactions result in the transfer of electrons from one atom to another. Oxidation is a loss of electrons and an increase in the number of carbon-heteroatom bonds; reduction is a gain of electrons and a decrease in the number of carbon-heteroatom bonds. In the electron transport chain, ubiquinone receives two electrons from NADH, which reduces the carbonyl carbons to hydroxyl groups and forms an aromatic ring, yielding ubiquinol.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 401469

System:

Functional Groups and Their Reactions